WEMU core – unit-test coverage report

Generated 2026-11-15 00:44 · commit `d9502bd` (branch `tests/quality-objectives`)

Scope	Lines	Functions	Target
Critical path (stages 1–6, core side)	82.5 % (16,240 / 19,675)	88.3 %	≥ 80 %: met
All WEMU core code	82.6 % (20,310 / 24,603)	80.8 %	≥ 70 %: met

Critical-path stage	Lines	Functions
1. Game loading (`binary/Loader.cpp`)	98.6 %	100 %
2. PowerPC execution (`cpu/interpreter`, `jit`, `instructions`, `memory`)	95.0 %	98.8 %
3. OS emulation (`Coreinit`, `CoreinitExtra`, `Fs`, `MemHeap`)	88.6 %	76.2 %
4. GX2 command processing (`hle/Gx2.cpp`, `gfx/Gx2Replayer.cpp`)	68.7 %	81.0 %
5. Shader translation and Vulkan	87.5 %	93.9 %
6. Presentation, core side (`video/`)	81.8 %	95.7 %

How it was measured: GCC 14 with `--coverage`, gcov-14, lcov 2.0; 1,305 GoogleTest cases run with `WEMU_TEST_VULKAN=1` (lavapipe). **Excluded:** `stb_image.h` (vendored), `core/src/gfx/init/` (needs a real window), `core/tests/`.

Contents: page 2 is the summary per directory; the following pages list every source file with its line and function coverage. The annotated source pages stay in the HTML report (`build-cov/coverage_report/`). Method, targets and the files below 80 %: `docs/quality/testing-policy.md` §3.

LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level	**Lines:**	82.6 %	24603	20310
Test:	cov-wemu.info	**Functions:**	80.8 %	2888	2333
Test Date:	2026-11-15 00:44:59				

Directory	Line Coverage			Function Coverage		
	Rate	Total	Hit	Rate	Total	Hit
audio/	90.4 %	437	395	97.6 %	41	40
binary/	97.4 %	384	374	88.9 %	36	32
cpu/instructions/	94.2 %	206	194	100.0 %	45	45
cpu/instructions/arithmetic/	98.0 %	151	148	88.0 %	25	22
cpu/instructions/float/	100.0 %	491	491	100.0 %	114	114
cpu/instructions/load/	100.0 %	107	107	100.0 %	21	21
cpu/instructions/logical/	100.0 %	58	58	100.0 %	16	16
cpu/instructions/shift/	100.0 %	57	57	100.0 %	9	9
cpu/instructions/spr/	100.0 %	60	60	100.0 %	13	13
cpu/instructions/store/	98.9 %	94	93	100.0 %	18	18
cpu/interpreter/	89.8 %	530	476	97.5 %	40	39
cpu/jit/	93.5 %	1812	1695	99.0 %	309	306
cpu/memory/	98.5 %	198	195	97.8 %	45	44
cpu/recompiler/	46.8 %	158	74	57.1 %	7	4
cpu/tables/	100.0 %	198	198	100.0 %	6	6
cpu/types/	92.9 %	14	13	100.0 %	5	5
exception/	83.3 %	6	5	60.0 %	5	3
gfx/	77.0 %	12939	9961	81.6 %	1001	817
hle/	85.6 %	5643	4831	83.1 %	706	587
ui/	96.2 %	156	150	100.0 %	21	21
utils/	81.1 %	618	501	39.0 %	382	149
video/	81.8 %	286	234	95.7 %	23	22

LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - audio	**Lines:**	**90.4 %**	437	395
Test:	cov-wemu.info	**Functions:**	**97.6 %**	41	40
Test Date:	2026-11-15 00:44:59				

File	Line Coverage			Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
MovieAudio.cpp		**86.5 %**	111	96	**100.0 %**	7	7
Output.cpp		**89.6 %**	211	189	**95.5 %**	22	21
Resampler.hpp		**93.3 %**	15	14	**100.0 %**	2	2
TimeStretcher.cpp		**97.4 %**	39	38	**100.0 %**	6	6
Voice.cpp		**95.1 %**	61	58	**100.0 %**	4	4

LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - binary	**Lines:**	**97.4 %**	384	374
Test:	cov-wemu.info	**Functions:**	**88.9 %**	36	32
Test Date:	2026-11-15 00:44:59				

File	Line Coverage			Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
Binary.hpp		**85.3 %**	34	29	**50.0 %**	6	3
Loader.cpp		**98.6 %**	347	342	**100.0 %**	26	26
Loader.hpp		**100.0 %**	3	3	**75.0 %**	4	3

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LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - cpu/instructions/arithmetic	**Lines:**	98.0 %	151	148
Test:	cov-wemu.info	**Functions:**	88.0 %	25	22
Test Date:	2026-11-15 00:44:59				

File	Line Coverage			Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
Add.cpp		95.0 %	60	57	75.0 %	12	9
Div.cpp		100.0 %	18	18	100.0 %	2	2
Mul.cpp		100.0 %	22	22	100.0 %	4	4
Neg.cpp		100.0 %	6	6	100.0 %	1	1
Sub.cpp		100.0 %	45	45	100.0 %	6	6

LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - cpu/instructions/float	**Lines:**	100.0 %	491	491
Test:	cov-wemu.info	**Functions:**	100.0 %	114	114
Test Date:	2026-11-15 00:44:59				

File	Line Coverage				Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
FloatArith.cpp		100.0 %	94	94	100.0 %	29	29
FloatCompare.cpp		100.0 %	47	47	100.0 %	3	3
FloatLoad.cpp		100.0 %	14	14	100.0 %	2	2
FloatLoadStoreExtra.cpp		100.0 %	75	75	100.0 %	18	18
FloatMove.cpp		100.0 %	12	12	100.0 %	3	3
PairedSingle.cpp		100.0 %	249	249	100.0 %	59	59

LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - cpu/instructions	**Lines:**	**94.2 %**	206	194
Test:	cov-wemu.info	**Functions:**	**100.0 %**	45	45
Test Date:	2026-11-15 00:44:59				

File	Line Coverage			Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
Branch.cpp		**96.0 %**	25	24	**100.0 %**	5	5
BranchExtra.cpp		**90.4 %**	73	66	**100.0 %**	4	4
Cmp.cpp		**100.0 %**	19	19	**100.0 %**	5	5
CrLogical.cpp		**100.0 %**	41	41	**100.0 %**	27	27
Sc.cpp		**91.9 %**	37	34	**100.0 %**	3	3
Trap.cpp		**90.9 %**	11	10	**100.0 %**	1	1

LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - cpu/instructions/load	**Lines:**	**100.0 %**	107	107
Test:	cov-wemu.info	**Functions:**	**100.0 %**	21	21
Test Date:	2026-11-15 00:44:59				

File	Line Coverage			Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
Load.cpp		**100.0 %**	53	53	**100.0 %**	11	11
LoadExtra.cpp		**100.0 %**	54	54	**100.0 %**	10	10

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LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - cpu/instructions/logical	**Lines:**	100.0 %	58	58
Test:	cov-wemu.info	**Functions:**	100.0 %	16	16
Test Date:	2026-11-15 00:44:59				

File	Line Coverage			Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
And.cpp		100.0 %	20	20	100.0 %	5	5
Ext.cpp		100.0 %	8	8	100.0 %	2	2
Or.cpp		100.0 %	17	17	100.0 %	5	5
Xor.cpp		100.0 %	13	13	100.0 %	4	4

LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - cpu/instructions/shift	**Lines:**	**100.0 %**	57	57
Test:	cov-wemu.info	**Functions:**	**100.0 %**	9	9
Test Date:	2026-11-15 00:44:59				

File	Line Coverage			Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
Extend.cpp		**100.0 %**	4	4	**100.0 %**	1	1
Rotate.cpp		**100.0 %**	25	25	**100.0 %**	4	4
Shift.cpp		**100.0 %**	28	28	**100.0 %**	4	4

LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - cpu/instructions/spr	**Lines:**	100.0 %	60	60
Test:	cov-wemu.info	**Functions:**	100.0 %	13	13
Test Date:	2026-11-15 00:44:59				

File	Line Coverage			Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
Cache.cpp		100.0 %	8	8	100.0 %	8	8
Spr.cpp		100.0 %	52	52	100.0 %	5	5

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LCOV - code coverage report

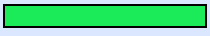
			Coverage	Total	Hit
Current view:	top level - cpu/instructions/store	**Lines:**	**98.9 %**	94	93
Test:	cov-wemu.info	**Functions:**	**100.0 %**	18	18
Test Date:	2026-11-15 00:44:59				

File	Line Coverage				Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
Store.cpp		**100.0 %**	39	39	**100.0 %**	8	8
StoreExtra.cpp		**98.2 %**	55	54	**100.0 %**	10	10

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LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - cpu/interpreter	**Lines:**	**89.8 %**	530	476
Test:	cov-wemu.info	**Functions:**	**97.5 %**	40	39
Test Date:	2026-11-15 00:44:59				

File	Line Coverage			Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
Interpreter.cpp		**88.8 %**	484	430	**100.0 %**	28	28
Interpreter.hpp		**100.0 %**	41	41	**90.0 %**	10	9
SyscallHandler.hpp		**100.0 %**	5	5	**100.0 %**	2	2

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LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - cpu/jit	**Lines:**	93.5 %	1812	1695
Test:	cov-wemu.info	**Functions:**	99.0 %	309	306
Test Date:	2026-11-15 00:44:59				

File	Line Coverage			Function Coverage		
	Rate	Total	Hit	Rate	Total	Hit
Callouts.cpp	77.1 %	35	27	99.5 %	195	194
CodeBuffer.cpp	86.4 %	44	38	100.0 %	6	6
CodeBuffer.hpp	100.0 %	4	4	100.0 %	4	4
EmitBranch.cpp	100.0 %	58	58	100.0 %	2	2
EmitFloat.cpp	99.8 %	430	429	100.0 %	20	20
EmitInteger.cpp	99.3 %	271	269	100.0 %	5	5
EmitMemory.cpp	100.0 %	187	187	100.0 %	8	8
EmitSystem.cpp	100.0 %	67	67	100.0 %	1	1
Emitter.cpp	92.3 %	414	382	100.0 %	31	31
Emitter.hpp	100.0 %	11	11	100.0 %	11	11
X64Jit.cpp	76.6 %	291	223	92.3 %	26	24

LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - cpu/memory	**Lines:**	98.5 %	198	195
Test:	cov-wemu.info	**Functions:**	97.8 %	45	44
Test Date:	2026-11-15 00:44:59				

File	Line Coverage				Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
Memory.hpp		98.5 %	198	195	97.8 %	45	44

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LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - cpu/recompiler	**Lines:**	46.8 %	158	74
Test:	cov-wemu.info	**Functions:**	57.1 %	7	4
Test Date:	2026-11-15 00:44:59				

File	Line Coverage				Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
Block.cpp		46.8 %	158	74	57.1 %	7	4

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LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - cpu/tables	**Lines:**	**100.0 %**	198	198
Test:	cov-wemu.info	**Functions:**	**100.0 %**	6	6
Test Date:	2026-11-15 00:44:59				

File	Line Coverage				Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
cpu_fields.anh		**100.0 %**	6	6	**100.0 %**	6	6
cpu_instructions.anh		**100.0 %**	192	192	**-**		

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LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - cpu/types	**Lines:**	92.9 %	14	13
Test:	cov-wemu.info	**Functions:**	100.0 %	5	5
Test Date:	2026-11-15 00:44:59				

File	Line Coverage				Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
EncodedInstruction.hpp		100.0 %	2	2	100.0 %	2	2
Instruction.hpp		66.7 %	3	2	100.0 %	1	1
InstructionTraits.hpp		100.0 %	9	9	100.0 %	2	2

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LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - exception	**Lines:**	83.3 %	6	5
Test:	cov-wemu.info	**Functions:**	60.0 %	5	3
Test Date:	2026-11-15 00:44:59				

File	Line Coverage				Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
Exception.cpp		100.0 %	1	1	50.0 %	2	1
Exception.hpp		80.0 %	5	4	66.7 %	3	2

LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - gfx	**Lines:**	77.0 %	12939	9961
Test:	cov-wemu.info	**Functions:**	81.6 %	1001	817
Test Date:	2026-11-15 00:44:59				

File	Line Coverage			Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
ColorTarget.hpp	71.4 %	105	75	100.0 %	12	12
CubeSampler.hpp	98.4 %	63	62	62.5 %	8	5
Depth.hpp	93.2 %	73	68	100.0 %	8	8
Gx2BufferSnapshots.hpp	82.5 %	80	66	100.0 %	3	3
Gx2CommandStream.hpp	97.3 %	75	73	97.1 %	34	33
Gx2Replayer.cpp	65.7 %	5612	3688	72.9 %	328	239
Gx2Replayer.hpp	96.1 %	77	74	97.1 %	34	33
HostPresenter.hpp	33.3 %	3	1	25.0 %	4	1
LatteProgram.cpp	95.5 %	111	106	100.0 %	2	2
LatteProgram.hpp	100.0 %	11	11	100.0 %	8	8
LatteShaderLowering.cpp	91.8 %	984	903	100.0 %	39	39
LatteShaderLowering.hpp	100.0 %	16	16	100.0 %	8	8
LatteSurfaceLayout.cpp	92.2 %	90	83	100.0 %	6	6
LatteVsInterp.cpp	85.6 %	728	623	95.8 %	24	23
PrimitiveAssembly.hpp	94.9 %	39	37	100.0 %	3	3
RasterBackend.hpp	71.6 %	67	48	51.9 %	27	14
RasterWorkers.cpp	86.6 %	67	58	100.0 %	8	8
Renderer.cpp	34.5 %	278	96	68.8 %	16	11
Renderer.hpp	42.9 %	28	12	80.0 %	10	8
SpirvCompiler.cpp	75.7 %	230	174	95.0 %	20	19
SpirvCompiler.hpp	100.0 %	8	8	100.0 %	4	4
SurfaceLayout.hpp	97.1 %	69	67	100.0 %	3	3
TextureFormat.hpp	100.0 %	34	34	100.0 %	5	5
TextureSampler.hpp	100.0 %	66	66	66.9 %	121	81
TriangleCoverage.hpp	100.0 %	18	18	100.0 %	3	3
UniformBufferFetch.hpp	92.3 %	13	12	100.0 %	1	1
VolumeSampler.hpp	100.0 %	6	6	33.3 %	9	3
VulkanPortability.hpp	96.3 %	27	26	100.0 %	6	6
VulkanRasterBackend.cpp	87.1 %	3952	3442	92.4 %	238	220
VulkanRasterBackend.hpp	88.9 %	9	8	88.9 %	9	8

LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - hle	**Lines:**	85.6 %	5643	4831
Test:	cov-wemu.info	**Functions:**	83.1 %	706	587
Test Date:	2026-11-15 00:44:59				

File	Line Coverage			Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
ApplicationBoot.cpp	96.9 %	32	31	100.0 %	2	2
AsyncCallbacks.cpp	74.1 %	85	63	90.0 %	10	9
Ax.cpp	87.9 %	694	610	84.1 %	69	58
Coreinit.cpp	82.4 %	905	746	62.2 %	156	97
CoreinitExtra.cpp	92.4 %	525	485	78.3 %	69	54
Fs.cpp	92.7 %	550	510	98.1 %	52	51
Gx2.cpp	81.4 %	1299	1057	89.2 %	166	148
H264.cpp	77.6 %	281	218	77.8 %	36	28
Libc.cpp	100.0 %	76	76	100.0 %	14	14
LockedCache.cpp	100.0 %	80	80	100.0 %	16	16
MemHeap.cpp	93.5 %	292	273	100.0 %	39	39
MvPlayer.cpp	66.2 %	231	153	92.3 %	13	12
NnOnline.cpp	87.9 %	107	94	66.7 %	12	8
Scheduler.cpp	88.4 %	441	390	97.4 %	38	37
Scheduler.hpp	100.0 %	12	12	100.0 %	7	7
SystemConfig.hpp	100.0 %	33	33	100.0 %	7	7

LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - ui	**Lines:**	**96.2 %**	156	150
Test:	cov-wemu.info	**Functions:**	**100.0 %**	21	21
Test Date:	2026-11-15 00:44:59				

File	Line Coverage				Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
PauseMenu.cpp		**95.8 %**	142	136	**100.0 %**	13	13
PauseMenu.hpp		**100.0 %**	14	14	**100.0 %**	8	8

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LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - utils	**Lines:**	81.1 %	618	501
Test:	cov-wemu.info	**Functions:**	39.0 %	382	149
Test Date:	2026-11-15 00:44:59				

File	Line Coverage			Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
BeDecoder.cpp		93.8 %	16	15	100.0 %	3	3
BeDecoder.hpp		96.6 %	29	28	91.7 %	12	11
Diagnostics.cpp		79.8 %	476	380	89.8 %	49	44
DisplayTiming.hpp		100.0 %	3	3	100.0 %	2	2
FrameProfiler.hpp		79.2 %	77	61	83.3 %	18	15
Logger.hpp		82.4 %	17	14	24.8 %	298	74

LCOV - code coverage report

			Coverage	Total	Hit
Current view:	top level - video	**Lines:**	81.8 %	286	234
Test:	cov-wemu.info	**Functions:**	95.7 %	23	22
Test Date:	2026-11-15 00:44:59				

File	Line Coverage			Function Coverage		
	Rate		Total	Hit	Rate	Total	Hit
H264Decoder.cpp		86.5 %	89	77	100.0 %	9	9
H264Sps.hpp		69.7 %	99	69	80.0 %	5	4
MovieDecoder.cpp		89.8 %	98	88	100.0 %	9	9